

A disproof of Conjecture 00000001756

Unimodular submatrices of the character table of S_3 : $d(3) = 2 \neq 3$

1 The conjecture and its reading

Conjecture 00000001756 reads: *Definition: A unimodular submatrix of the character table of S_n is a square submatrix with determinant ± 1 . Conjecture: Its maximal order $d(n)$ is characterized by the hook-coprimality condition: $d(n) = \#\{\lambda \vdash n : \text{hook}(\lambda) \text{ pairwise coprime}\}$, an explicit formula; a submatrix of that order can be selected from irreducible-character rows (sorted by dimension).*

The conjecture is a conjunction. Its first clause asserts the identity $d(n) = \#\{\lambda \vdash n : \text{the hook lengths of } \lambda \text{ are pairwise coprime}\}$ for every n . We show that this identity fails at $n = 3$, where $d(3) = 2$ and the right side is 3. The remaining clauses are not used.

A submatrix is given by a choice of k distinct rows and k distinct columns, which we allow in any order. Reordering rows or columns changes the determinant only by a sign, so this does not affect $d(n)$.

2 The right-hand side: hook lengths of the partitions of 3

Definition 1. A partition $\lambda \vdash n$ is a multiset of positive integers with sum n (Mathlib's `Nat.Partition n`). Its Ferrers diagram is the set of cells (i, j) such that at least $i + 1$ parts are $\geq j + 1$. It is a Mathlib `YoungDiagram`. The hook length of a cell $c = (i, j)$ is

$$h(c) = \#\{\text{cells to the right of } c \text{ in row } i\} + \#\{\text{cells below } c \text{ in column } j\} + 1,$$

that is, arm + leg + 1. The hook lengths of λ are *pairwise coprime* if $\gcd(h(c), h(c')) = 1$ for every two distinct cells $c \neq c'$.

Lemma 2. *The partitions of 3 are exactly (3) , $(2, 1)$ and $(1, 1, 1)$. Their Ferrers diagrams have cells $\{(0, 0), (0, 1), (0, 2)\}$, $\{(0, 0), (0, 1), (1, 0)\}$ and $\{(0, 0), (1, 0), (2, 0)\}$. Their hook lengths are 3, 2, 1, then 3, 1, 1, then 3, 2, 1. All three have pairwise coprime hook lengths. Hence the right side equals 3.*

Proof. A multiset of positive integers with sum 3 is $\{3\}$, $\{2, 1\}$ or $\{1, 1, 1\}$. The cells and hook lengths are a finite computation, and every pair among $\{3, 2, 1\}$ or $\{3, 1, 1\}$ is coprime. $\square$

Remark 3. The value 3 does not depend on how “hook(λ) coprime” is read. Every reading implied by pairwise coprimality over distinct cells gives 3, because every partition of 3 satisfies that strongest condition. This covers pairwise coprimality of the distinct hook values and $\gcd$ of all hooks = 1.

3 The left-hand side: $d(3) = 2$

Let $G = S_3$ be the permutation group of $\{0, 1, 2\}$. All representations are finite-dimensional complex representations of G .

Definition 4 (Character table). A *character table* of G is a matrix Y with entries $Y_{ij} = \chi_{V_i}(g_j)$, where $V_1, \dots, V_m$ is a list of irreducible representations containing every irreducible representation exactly once up to isomorphism, and $g_1, \dots, g_{m'}$ is a list of elements meeting every conjugacy class exactly once. Both lists may be in any order, and m, m' are not fixed in advance. In Lean this is the predicate `IsCharTable`. Irreducibility is Mathlib's `Simple` in the category `FDRep ℂ` (`Perm (Fin 3)`), and χ_V is Mathlib's `FDRep.character`, the trace of $\rho_V(g)$.

Lemma 5 (The irreducible representations of S_3). *Let ρ_0 be the trivial representation on $\mathbb{C}$ and $\rho_1 = \text{sgn}$ on $\mathbb{C}$. Let ρ_2 be the action of G on the plane $P = \{x \in \mathbb{C}^3 : x_0 + x_1 + x_2 = 0\}$ by permuting coordinates. We write it in the basis $f_j = e_j - e_2$ ($j = 0, 1$); a vector of P has coordinates (x_0, x_1) in this basis. So column j of $\rho_2(\sigma)$ is $e_{\sigma(j)} - e_{\sigma(2)}$, restricted to its first two coordinates. Then:*

1. the characters of ρ_0, ρ_1, ρ_2 are $\chi_0 = 1$, $\chi_1 = \text{sgn}$ and $\chi_2(\sigma) = \#\text{Fix}(\sigma) - 1$;
2. ρ_0, ρ_1, ρ_2 are irreducible and pairwise non-isomorphic;
3. every irreducible representation of G is isomorphic to one of them.

Proof. The three maps $\sigma \mapsto \rho_i(\sigma)$ are integer matrices. Lean checks the homomorphism property and computes the traces over the six elements of G , then transports the matrices to $\mathbb{C}$ by `Matrix.toLinAlgEquiv`.

(2) Over an algebraically closed field of characteristic 0, a representation V is irreducible if and only if $\sum_g \chi_V(g) \chi_V(g^{-1}) = |G|$. This is Mathlib's `FDRep.simple_iff_char_is_norm_one`. For $i = 0, 1, 2$ the sum equals 6. Isomorphic representations have equal characters (`FDRep.char_iso`), and the three characters are pairwise different.

(3) Let V be irreducible and assume it is isomorphic to none of the ρ_i . By the orthogonality relations (Mathlib's `FDRep.char_orthonormal`), $\sum_g \chi_V(g) \chi_i(g^{-1}) = 0$ for $i = 0, 1, 2$. The character χ_V is a class function. Let a, b, c be its values on `id`, `(0 1)` and `(0 1 2)`. The classes have sizes 1, 3, 2, so

$$a + 3b + 2c = 0, \quad a - 3b + 2c = 0, \quad 2a - 2c = 0.$$

Hence $a = b = c = 0$ and $\chi_V = 0$. This contradicts $\sum_g \chi_V(g) \chi_V(g^{-1}) = 6$, which holds by (2) applied to V . $\square$

Corollary 6. *Let*

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & -1 \end{pmatrix},$$

the table of χ_0, χ_1, χ_2 at `id`, `(0 1)`, `(0 1 2)`. Every character table Y of G satisfies $Y_{ij} = X_{\pi(i)\tau(j)}$ for some bijections $\pi : \{1..m\} \rightarrow \{0, 1, 2\}$ and $\tau : \{1..m'\} \rightarrow \{0, 1, 2\}$. In particular, the table built from ρ_0, ρ_1, ρ_2 and the three representatives is a character table, and it equals X .

Proof. By Lemma 5(3), $V_i \cong \rho_{\pi(i)}$ for some $\pi(i)$. The map π is injective because the V_i are pairwise non-isomorphic. It is surjective because each ρ_p is irreducible, so it is isomorphic to some V_i , and then $p = \pi(i)$ by Lemma 5(2). Let $\tau(j)$ be the class of g_j . Then τ is a bijection because the g_j meet every class exactly once. Finally, $\chi_{V_i}(g_j) = \chi_{\pi(i)}(g_j) = X_{\pi(i)\tau(j)}$, since characters of isomorphic representations agree and characters are class functions. $\square$

Theorem 7. *Every character table Y of S_3 has $d(Y) = 2$. In particular $d(3) = 2$.*

Proof. By Corollary 6, rows r and columns c of Y give the minor of X on rows $\pi \circ r$ and columns $\tau \circ c$, cast to $\mathbb{C}$. Conversely, rows r and columns c of X arise from $\pi^{-1} \circ r$ and $\tau^{-1} \circ c$. An integer equals ± 1 exactly when its complex cast does. So Y and X have the same unimodular orders, and it suffices to treat X .

Rows $\{\chi_0, \chi_2\}$ and columns $\{(0\ 1), (0\ 1\ 2)\}$ give $\det \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = -1$, so $d(X) \geq 2$. A 3×3 submatrix uses all rows and all columns in some order. Its determinant is therefore $\text{sgn}(e) \text{sgn}(f) \det X = \pm 6$, where e, f are the row and column permutations and $\det X = 6$. No square submatrix has order ≥ 4 . $\square$

Theorem 8 (Main theorem). *For every character table of S_3 , $d(3) = 2$, and $\#\{\lambda \vdash 3 : \text{hooks pairwise coprime}\} = 3$. Hence $d(3) \neq 3$ and the conjectured formula fails.*

4 Relation to the earlier submission

PR #214 by orionsheep was closed unmerged. The reviewer accepted the mathematics. The reason for closing was: “the capstone theorem `conjecture_refuted` : $\neg((3 : \text{Nat}) = 2)$ connects none of them: the refutation is assembled only in prose. Happy to re-review a version whose main theorem states $d(3) = 2 \neq 3$ on the actual hook/partition objects.”

The present main theorem `conjecture_1756_false` states $\text{d3} = 2 \wedge \text{hookCount } 3 = 3 \wedge \text{d3} \neq \text{hookCount } 3$. Here d3 is the supremum of the orders of ± 1 -determinant square submatrices of the character table built in Lean from the irreducible representations of Lemma 5. The companion theorem `conjecture_1756_false_any_table` gives the same value for every character table, in any order of rows and columns. `hookCount n` counts Mathlib partitions of n whose Ferrers YoungDiagram has pairwise coprime hook lengths. The hook lengths are computed from the cells, not hardcoded.

5 Lean formalization

The file is `lean/Conjecture1756/Basic.lean` (namespace `C1756`), and it imports only Mathlib. The Ferrers-diagram construction (`rowsAtLeast`, `ferrers`) is adapted, with credit, from the accepted solution of conjecture 00000000428 by ColdSmile, which is under GPL-3.0.

- Partitions and hooks:
 - definitions `ferrers`, `hookLength`, `HookCoprime`, `hookCount`;
 - `partitions_three`: every partition of 3 is (3), (2, 1) or (1, 1, 1);
 - `ferrers_P3`, `ferrers_P21`, `ferrers_P111`: the diagrams;
 - `hooks_P3`, `hooks_P21`, `hooks_P111`: the hook lengths;
 - `hookCount_three`: the count is 3.
- Irreducible representations (Lemma 5):
 - `matRep`: an integer matrix homomorphism $G \rightarrow M_n(\mathbb{Z})$ as an object of `FDRep` $\mathbb{C}$ G ; `trivHom`, `signHom`, `stdHom` and `irr`;
 - `irr_character`: the characters are `chi 0`, `chi 1`, `chi 2`;

- `irr_simple`: each `irr i` is `Simple`;
 - `irr_distinct`: they are pairwise non-isomorphic;
 - `irr_complete`: every `Simple` object of `FDRep  $\mathbb{C}$  G` is isomorphic to some `irr i`.
- Character tables (Corollary 6):
 - `IsCharTable`: the predicate of Definition 4;
 - `IsCharTable.reindex`: every character table is `charTable` with rows and columns permuted;
 - `trueTable`, `isCharTable_trueTable`, `trueTable_eq`: the table of `irr` at `rep` is a character table, and it equals X .
 - Unimodular orders (Theorem 7):
 - definitions `IsUnimodularOrder`, `maxUnimodularOrder` (for rectangular matrices over any commutative ring) and `d3 := maxUnimodularOrder trueTable`;
 - `det_full_submatrix`: every 3×3 submatrix of X has determinant ± 6 ;
 - `unimodular_two`, `unimodular_le_two`: the unimodular orders of X ;
 - `isUnimodularOrder_reindex`: permuting and casting to $\mathbb{C}$ preserves them;
 - `maxUnimodularOrder_of_isCharTable`: every character table has maximal unimodular order 2; `d3_eq_two`.
 - Main theorems: `conjecture_1756_false : d3 = 2  $\wedge$  hookCount 3 = 3  $\wedge$  d3  $\neq$  hookCount 3`, and `conjecture_1756_false_any_table`, the same statement for every X with `IsCharTable X`.
 - Axiom audit: `#print axioms` on both main theorems reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`. The `decide` calls run over the finite types `Perm (Fin 3)`, `Fin 3`, small integer matrices and explicit finite cell sets.